

Salvatore Barbagallo

Bioinformatics Analyst | NGS & Computational Genomics

London, United Kingdom

+44 (0)7475200048 | barbagallo1@hotmail.it | [LinkedIn](#) | [Portfolio](#)

Professional Summary

Bioinformatics and data professional with 7+ years of experience in clinical genomics, specialising in building reproducible NGS pipelines and scalable data workflows.

Designs and implements end-to-end RNA-seq and variant analysis systems using Python, R, Bash, and SQL, with a strong focus on data quality, automation, and workflow reliability in regulated environments. Combines laboratory expertise with computational engineering to deliver robust, decision-ready genomic analyses.

Skills

Programming: Python, R, SQL, Bash, Git

Pipelines & Infrastructure: Nextflow, Docker, Galaxy, AWS, workflow automation

NGS & Omics: RNA-seq, WGS, variant calling, QC pipelines, STAR, BWA-MEM2, bcftools, SnpEff, DESeq2, IGV

ML & Data: scikit-learn, XGBoost, PyTorch, Keras, BigQuery, Tableau

Languages: Italian (Native), English (Fluent), Portuguese (Fluent), Spanish (Advanced)

Professional Experience

Bioinformatician - Computational STEM Masters

Mercor, Contract, January 2026 – July 2026

- Authored 106 hidden-function tasks accepted into a frontier model's bioinformatics benchmark, each requiring the model to reverse-engineer a novel function by feeding it inputs and reading the outputs.
- Calibrated task difficulty against measured model-performance curves using ablation ladders and adversarial foils, holding textbook solutions below 50% while the full solution reached 100%, at 80% first-pass approval through peer and client review.

Specialist Biomedical Scientist

University College London Hospitals, Stem Cell Laboratory, *September 2021 – December 2025*

- Designed and implemented automated tracking tools (Python/Excel) to replace manual reconciliation, reducing inventory errors by ~30% and saving ~10 hours per week.
- Supported 11 clinical and ATMP trials including CAR-T therapies, ensuring compliance and high-quality data handling; processed PBSC, bone marrow, DLI and CD34+ products with flow cytometry analysis of CD3+/CD34+ populations.

Laboratory Scientist

CooperGenomics, London, *July 2019 – September 2021*

- Processed and quality-controlled 500+ NGS samples per week (library prep, 96–192 samples/run) for PGT-A, PGT-SR and PGT-M, delivering 500+ validated clinical reports weekly against turnaround targets.
- Programmed and validated liquid-handling robots (Mosquito HV, Dragonfly) and authored SOPs to ISO-compliant standards, enabling scalable, reproducible workflows.

Projects

Causal Assessment of GPNMB in Parkinson's Disease (R, TwoSampleMR, coloc, GWAS, pQTL/eQTL)

- Designed and executed a two-sample Mendelian randomisation and Bayesian colocalisation workflow to evaluate causal evidence for a therapeutic target using human genetic summary data.
- Harmonised exposure and outcome data, evaluated instrument strength and linkage disequilibrium, generated effect estimates and confidence intervals, and performed sensitivity analyses for pleiotropy, assay artefact and cohort provenance.

Reproducible RNA-seq Pipeline (Nextflow, Docker)

- Built a containerised RNA-seq pipeline (Nextflow, Docker) to eliminate environment inconsistencies and manual execution steps, enabling reproducible one-command analysis from raw FASTQ to differential expression results.
- Integrated QC, alignment, quantification, and statistical analysis into a scalable workflow suitable for research and clinical genomics contexts.

End-to-End WGS Pipeline for Drug Resistance Analysis (Galaxy, Linux, bcftools)

- Developed an end-to-end WGS pipeline for Mycobacterium tuberculosis to identify drug resistance mutations, integrating QC, alignment, coverage analysis, variant calling, and annotation into a reproducible Galaxy workflow.
- Enabled clinically relevant interpretation of resistance-associated loci using allele frequency, coverage metrics, and IGV-based validation, supporting interpretation of clinically relevant resistance mutations in infectious disease analysis.

Education

MSc Bioinformatics – *Atlantic Technological University, September 2025 - September 2026*

- *Dissertation: Prioritising causal genes for autoimmune disease using Mendelian randomisation and colocalisation.*

MSc Cell and Gene Therapy – *University College London, September 2021- September 2023*

- *Dissertation: Expansion and Preservation of Haematopoietic Potential in Human Amniotic Fluid Stem Cells for Therapeutic Applications*

BSc Biomedical Science – *University of Catania, September 2014 - October 2017*

Certificates

Ulster University: Coding Skills for Biologists

IBM: Machine Learning; AI Engineering; Data Engineering

Google: Data Analytics; Advanced Data Analytics; IT Automation with Python; Project Management; Business Intelligence

Google Cloud: Architecting with Google Kubernetes Engine

Amazon Web Services (AWS): Cloud Practitioner Essentials; Cloud Solutions Architect

Johns Hopkins University: Genomic Data Science Specialization

SAS: SAS Programming 1: Essentials; SAS Programming 2: Data Manipulation Techniques